

Research Briefing | EU

Unusual cold spell will weigh on Q1 GDP

- **The coldest start to a year in nearly a decade presents downside risk to our Q1 GDP forecasts for the EU. A decline in EU GDP isn't very likely. But falls in construction investment worth between 0.2%-1% of GDP in Germany, Poland, and the Czech Republic could drag these economies into contraction and would create alone a 0.1ppt drag on EU GDP growth.**
- **A larger drag on EU growth is possible. Construction is likely to disappoint in other countries as well. Hospitality spending may be soft, while stronger energy demand will be a partial offset. February surveys and January activity data will give initial indications whether Q1 GDP will indeed disappoint, which will inform any upcoming forecast changes.**
- **A significant drag from the cold spell on Q1 GDP would support our below-consensus growth forecast for the Eurozone for this year, as Q1 shortfalls don't tend to be recouped swiftly. However, given the temporary nature of this impact, we see no policy implications.**

Europe has experienced the coldest start to a year in nearly a decade. This had already contributed to spikes in natural gas prices earlier in the year given the unusually high demand for heating energy. While the lift to gas prices was short-lived, the real economic fallout will become apparent over the next few weeks.

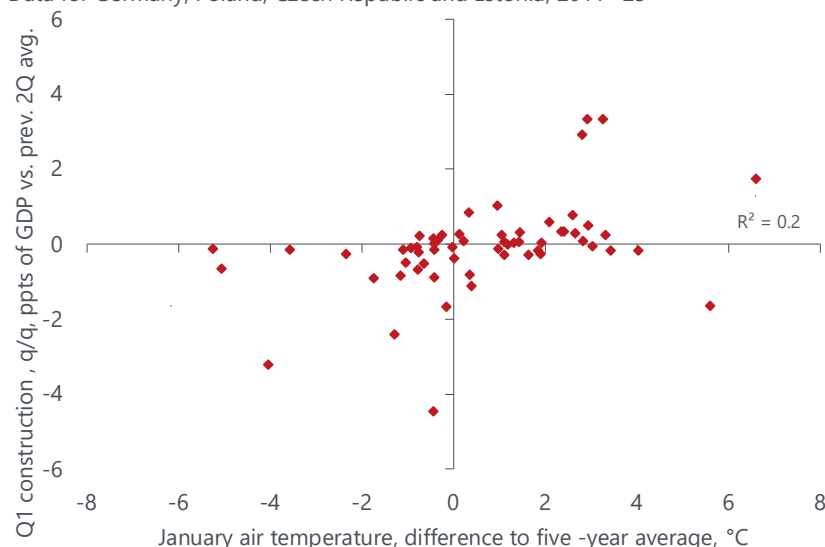
We estimate the cold spell has generated a drag of at least 0.1ppt on Q1 GDP growth in the EU, with adverse effects concentrated in construction in central and eastern EU countries such as Germany, Poland, and the Czech Republic (**Chart 1**). This is evidenced by January's 10% m/m plunge in Polish construction activity, marking the second-worst reading in at least 15 years.

Risks skew to a larger growth hit given the material undershoot of recent temperature trends across a larger number of EU countries and the extension of the cold spell well into February. Moreover, there are indications that countries in southern Europe not significantly affected by the cold spell are experiencing other adverse weather events, such as unusually strong and lasting storms that are weighing on activity.

Chart 1: Construction tends to drag on Q1 GDP when temperatures fall below recent norms

EU: January temperature and Q1 construction investment

Data for Germany, Poland, Czech Republic and Estonia, 2011 -25



Sources: Oxford Economics, Haver Analytics, Our World in Data

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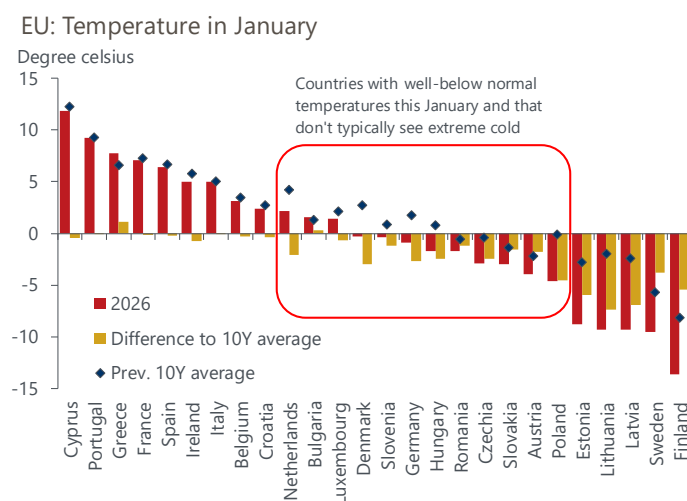
Details from surveys as well as the first batch of hard data for January will help to inform possible downgrades to our Q1 GDP forecasts over the next weeks. We expect monthly construction data will bear the brunt of the hit, but certain consumer services may also be adversely affected. On the other hand, energy production will benefit, while the response of retail sales is uncertain.

Central and Eastern Europe have been hit the most by the recent cold spell

The cold spell has predominantly affected central, northern, and eastern EU countries. Temperatures in January this year materially undershot recent historical averages, leaving them below zero in most countries (**Chart 2**). The GDP-weighted average temperature across EU countries fell to a nine-year low. This is also true for Germany and the Czech Republic. Temperatures in Poland and Estonia dropped even further to a 16-year low.

The undershoot is even more stark when we consider the trends over the past five years instead of ten years, given the general uptrend in temperatures over recent decades. This is important because seasonal adjustment for Q1 activity will have gradually accounted for a lesser "normal" drag from winter weather due to the trend increase in temperature. So, a break in that trend could amplify the impact on the activity data we typically analyse.

Chart 2: Much of Europe has experienced a cold spell this year



Sources: Oxford Economics, Our World in Data

It is also noteworthy that in many of the countries affected by the cold spell, temperatures don't typically fall much below zero, which is a crucial threshold for many major construction activities. Temperatures a few more degrees below zero than normal in some countries, such as Finland, won't make much of a difference to construction and other outside activities, and could even lift demand for heating above normal. However, a cold January and February could easily delay projects by a few months in other countries, such as Germany, that are more temperate, especially in recent years. For example, a large [train track project](#) in northern Germany has already been affected.

Construction is set to be the most affected sector by the cold spell

Results from our regression analysis across different timeframes tally well with the above logic. We find the most compelling evidence of a significant temperature effect on construction activity in Q1 for Germany, Poland, the Czech Republic, and Estonia (**Chart 1**). In 80% of the countries where the temperature in January fell below its five-year average, construction was a drag on Q1 GDP growth.

We think a drag worth 0.2ppts of GDP in Q1 in Germany and the Czech Republic seems feasible based just on the January temperature data. Meanwhile, the drag could amount to up to 1ppt in Poland given that temperatures are twice as far below "normal" as in the other two countries. The reason it tends to suffice to look at January is that even if temperatures quickly normalize in the rest of the quarter, firms will struggle

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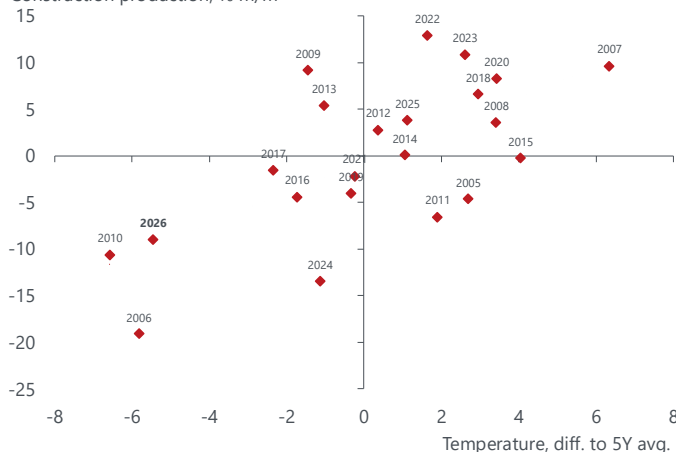
to make up the shortfall quickly. Either way, temperatures stayed lower than usual in February in these countries.

There is already early evidence of a hit to construction. As mentioned, activity in Poland dropped sharply in January, down by close to 10% m/m, the second-worst reading in at least 15 years (**Chart 3**). Historically, such large falls have nearly always resulted in a Q1 contraction in construction activity.

Chart 3: Hard data for Poland confirm the Q1 drag from weather on construction

Poland: Construction and temperatures in January

Construction production, % m/m



Sources: Oxford Economics, Haver Analytics, Our World in Data

In other countries, the historical evidence of Q1 construction activity being sensitive to unusually cold weather is less clear. But this may have to do more with data issues rather than lack of an impact. So, we think it's likely that construction will suffer a false start to the year in more countries than just the four highlighted above.

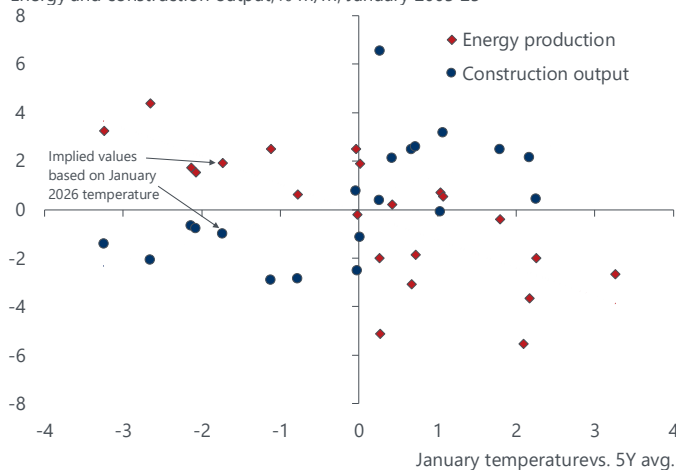
Weather impact in Q1 is likely to be more mixed outside of construction

Other parts of the economy may experience a temporary boost from this year's cold spell. For instance, there is an obviously close relationship between energy demand and temperatures. The very cold January may have lifted energy output in the EU by around 2% in January, while the impact on construction at least in aggregate may have been a little smaller (**Chart 4**). This means that the impact on January activity data is not unambiguously negative at the EU or Eurozone level.

Chart 4: Higher energy demand in response to the cold spell will partially offset construction declines

EU: Energy and construction output versus temperatures

Energy and construction output, % m/m, January 2005-25



Sources: Oxford Economics, Haver Analytics, OWID

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However, we think aggregate data are likely to understate the impact given the large country-specific falls implied by the data. Moreover, the construction sector is five times the size of the energy sector, when looking at gross value added. So, even accounting for this offset, we would expect a net drag on Q1 GDP.

Even if the lift from higher energy demand was more pronounced than we think, history suggests that a cold winter is a net drag on services activity. The evidence is less clear, especially in recent years, as seasonal adjustment has struggled to account for the volatility during and after the coronavirus pandemic. But restricting the analysis to pre-pandemic years, a fall in hospitality turnover between 1%-2% m/m in January looks feasible, partially offset by higher retail sales as people consume more at home.

In addition, there is some evidence that adverse weather has also affected southern European countries. For instance, in Spain, [high-frequency data](#) suggest consumer spending sharply lost momentum at the start of this year. Similarly, in [Portugal](#), we expect a soft Q1 GDP reading partly driven by recent severe weather events that disrupted production, with the extensive physical damage to factories and infrastructure, including the country's most important highway, adding to the hit.

Data to watch

A noticeable hit to Q1 GDP is not a foregone conclusion, though. Therefore, we do not plan on pre-emptively cutting our near-term forecasts. Surveys such as the [PMIs](#) covering services and industry have had a robust start to this year, with a solid rise in February reversing the softness from January. Given the relative sizes of these sectors, their gains may outweigh the weakness signalled in some [construction surveys](#). Nonetheless, risks are clearly tilted to the downside for Q1 mostly in central and eastern Europe, so we will watch incoming data to gauge possible forecast changes.

The next major data point is the February ESI survey later this week, as it includes a question on the share of construction firms suffering under adverse weather, which has historically often aligned with output changes. In early March, the release of the February construction PMIs and German industrial and construction production will also be crucial. January hard data for the EU and Eurozone will only be available by mid- to late-March, with industrial production due March 13 and construction output due March 19. National activity data is usually released after the German but before the EU data.

Broader implications

We see three broader takeaways. First, our baseline is well positioned for a significant drag from the cold spell on Q1 GDP growth given our below-consensus projections for Q1 and H1 GDP growth. While the improvement in several surveys had recently signalled emerging upside risks to our forecast, the cold spell at least partially negates them. Importantly, while activity is set to bounce back in Q2, Q1 losses aren't usually fully made up in Q2.

Second, we don't see any implication for our baseline views on European Central Bank policy. Even if growth undershoots the ECB baseline in H1, a temporary weather-induced growth dent cannot be addressed by monetary policy. This is the case even if the drag on Q1 GDP should be larger than the 0.1ppt-0.2ppts that we think are feasible.

Third, the likely volatility in hard data in H1 will make it difficult to ascertain the underlying growth trend in the next months, complicating the ECB's job. It will be easy for more hawkish council members to dismiss any softness, blame the weather, and play for time. So, we don't think that a moderate undershoot of the ECB's robust baseline GDP forecast in H1 would suffice to trigger a policy response. Instead, we think that any policy response is more likely to be triggered by softer-than-expected inflation figures. Inflation tends to be less affected by weather-related volatility.